

System Design:

The challenge has moved from designing a board to its fabrication

Today, the design trend is moving from designing printed circuit boards (PCBs) to designing a complete system, be it a car automation system, medical automation system or phone. Let us take a look at the changes starting at the chip level. Saugat Sen, vice president, research and development, Silicon-Package-Board R&D Group, Cadence, talks to Priya Ravindran of EFY about PCB designing and its challenges



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Q. What trends have you observed in PCB designing?

A. With advances in chip technology, the process of package design is becoming complex. There is a need for electrical analysis at the package level and board level to be comprehended together for a design to be successful. Speeds of signals across the interconnect are so fast that routing technologies have to be able to take electrical effects into account. Also, design tools are becoming more user-friendly to achieve faster time-to-market.

Q. How is PCB designing different from chip designing?

A. PCB designing and chip designing have parallels. While the front-end of chip designing includes the register transfer level creation, simulation and synthesis stage, front-end of PCB designing involves schematic creation and simulation. The schematic stage deals with choosing components and deciding the interconnection between these and between different boards. This is followed by simulation for analogue signals. Once this is done, the layout is created where the physical aspects of the PCB come into play.

Q. Would the approach for analogue, digital and mixed signals be different when it comes to PCB designing?

A. When you consider system design in totality, the final design is multi-disciplinary and involves interconnection of boards. While designing, we have to comply with electrical disciplines. For example, in radio frequency designs, analogue is more of a simulation-and-analysis challenge and digital involves dealing with high-speed signals. A strong interconnect engine is required, and the placement and routing stage of PCB designing transcends all three disciplines.

Interconnect engines would be in terms of standard interfaces and protocols. It is important to comprehend these protocols completely so the customer would not have to place and route the signals line by line but can work with the protocol as a whole to interconnect devices.

Q. What is the role of design reuse in the PCB design process, and how feasible is it?

A. There are many iterations that happen between the front-end and the back-end of board designing, that is, between the schematic and the layout stage. A streamlined

front-to-back and back-to-front flow is important for faster time-to-market. Most designs reuse portions of previously-designed PCBs within the same product architecture. Reuse involves reusing schematics, layout and constraints that go along with these.

Q. What does the verification process of PCB designing involve on the analysis front?

A. You simulate the circuit and the results seem satisfactory, but when you translate these to a hardware design, results may not be the same. Especially with high-speed signals, we find that from the start to the end, signal integrity is not maintained. It undergoes noise and reflection, thus making sure that maintaining signal integrity becomes very important. There is also the issue of power levels along the board, so power-aware-signal-integrity analysis is the way forward.

Q. What are the challenges in multi-layer board designing?

A. With multi-layer technology, the challenge has moved from designing a board to its fabrication. To come up with new fabrication technologies that can build high-density PCBs, with the necessary components and vias embedded between layers, is the task of the day. Also, analysis of multi-layer boards, from electrical and thermal perspectives, introduces its own challenges.

Q. How would you compare open source tools with proprietary ones?

A. Open source tools are great for learning and provide a platform and scope for knowledge and improvement. For commercial purposes, proprietary tools might suit better, simply because of the amount of investment, in terms of money, time and responsibility.

Q. Have your design tools been used for any interesting projects?

A. Rigidflex designs or flexible electronics is one interesting area where Allegro has been used. Our tools have been used for building medical devices such as miniscule hearing aids, for designing drill bits to search for underground oil and in building power supplies for satellites and space probes. ●